

Why the sphere beats the ball

Vamshi Jandhyala

Abstract

Three random points on a circle form an acute triangle less often than three random points in the disc, but three random points on a sphere do so more often than three in the ball. We explain the second half exactly. For three points with independent uniform directions on concentric spheres of radii $a \geq b \geq c > 0$ we find the probability of an acute triangle in closed form and show that it is at most $\frac{1}{2}$, with equality only when the two largest radii are equal. Consequently the uniform sphere is the unique best rotationally invariant distribution in space, and the ball must lose to it. In the plane a small amount of mass at the centre helps instead, which is where the two dimensions part ways. The algebra and the integrals are checked in the Lean proof assistant.

1 The question

Choose three independent uniform points from a set and ask how often they form an acute triangle. Dan asked on MathOverflow [3] why the four classical answers behave as they do:

	boundary	interior
plane	circle: $\frac{1}{4}$	disc: $\frac{4}{\pi^2} - \frac{1}{8} \approx 0.280$
space	sphere: $\frac{1}{2}$	ball: $\frac{33}{70} \approx 0.471$

The disc and ball values are Hall's [1]; the circle value is classical. Filling in the circle helps, but filling in the sphere hurts. Here is a version to try before reading on. Three points on a sphere give an acute triangle half the time. Now pull one of them inside, to a smaller sphere with the same centre, keeping its direction uniform. Does the chance go up or down?

It goes down, and the same happens whatever the three radii are, unless the two largest agree.

Theorem 1. *Let A, B, C be independent, uniform on the spheres of radii $a \geq b \geq c > 0$ centred at the origin of $\mathbb{R}^3$, and put $h = \sqrt{\max(0, b^2 + c^2 - a^2)}$. Then*

$$P(a, b, c) = \Pr(ABC \text{ is acute}) = \frac{b}{2a} + \frac{c^2}{6ab} - \frac{h^3}{6abc} \leq \frac{1}{2},$$

with equality if and only if $a = b$.

Read the formula with $a = b = c$: then $h = c$ and $P = \frac{1}{2} + \frac{1}{6} - \frac{1}{6}$, the sphere. Moving the outer point further out (a larger) costs at once (Figure 1, left), and moving the inner point in costs nothing only while the other two share a sphere.

Corollary 1. *Let X, Y, Z be independent with a common rotationally invariant distribution on $\mathbb{R}^3$. Then $\Pr(XYZ \text{ is acute}) \leq \frac{1}{2}$, with equality if and only if the distribution is uniform on a sphere centred at the origin. In particular the ball, whose radius is spread out, does worse than the sphere, whatever its exact value.*

The plane behaves differently, and a two-line experiment shows it (Section 5): put a small mass ε at the centre of the circle and the probability rises, while the same change to the sphere makes it fall (Figure 1, right). What we do not have is a direct proof that the disc's particular spread helps; its value is only known by computing it.

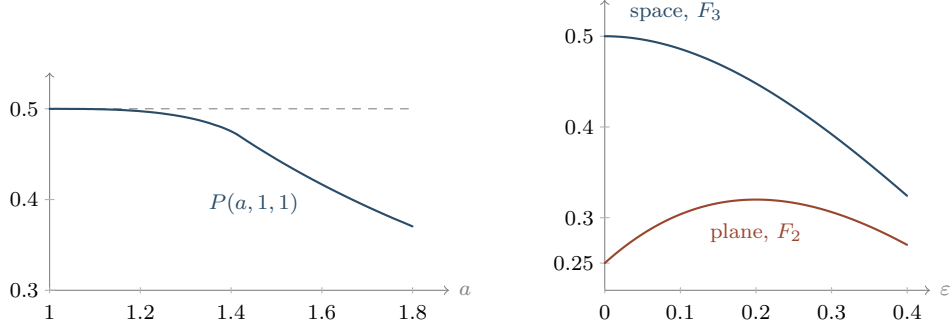


Figure 1: Left: $P(a, 1, 1)$ from Theorem 1, one point on a sphere of radius $a \geq 1$ and two on the unit sphere; it starts at $\frac{1}{2}$ and falls. Right: the centre experiment of Section 5, a point at the centre with probability ϵ , otherwise uniform on the circle or sphere.

2 Caps on a sphere

Everything reduces to one-dimensional integrals because of Archimedes' hat-box theorem: if W is uniform on a sphere of radius w in $\mathbb{R}^3$, its projection onto any fixed line through the centre is uniform on $[-w, w]$.

Lemma 1. *The angle of triangle UVW at W is obtuse if and only if $(U+V) \cdot W > U \cdot V + |W|^2$.*

Proof. The angle at W is obtuse when $(U - W) \cdot (V - W) < 0$, and expanding gives $U \cdot V - (U + V) \cdot W + |W|^2 < 0$. $\square$

So once U and V are fixed, the bad positions of W form a cap: the part of its sphere beyond a plane perpendicular to $U + V$ (Figure 2).

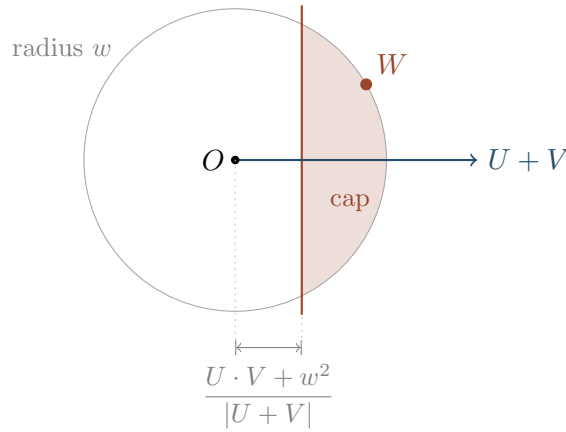


Figure 2: With U and V fixed, the angle at W is obtuse exactly when W lies in the shaded cap, beyond the plane at height $(U \cdot V + w^2)/|U + V|$ along $U + V$.

Let u, v, w be the radii of U, V, W and $M = |U + V|$. By the hat-box theorem $U \cdot V/(uv)$ is uniform on $[-1, 1]$, so from $M^2 = u^2 + v^2 + 2U \cdot V$ the length M has density $M/(2uv)$ on $[|u - v|, u + v]$. Given U and V , the projection $(U + V) \cdot W$ is uniform on $[-wM, wM]$, so it exceeds the threshold $T = U \cdot V + w^2 = \frac{1}{2}(M^2 + w^2 - K)$, with $K = u^2 + v^2 - w^2$, with probability $(wM - T)/(2wM)$ clipped to $[0, 1]$. Simplifying, $(wM - T)/(2wM) = (K - (M - w)^2)/(4wM)$, and the probability of an obtuse angle at W is

$$p_W = \int_{|u-v|}^{u+v} \max\left(0, \min\left(\frac{M}{2uv}, \frac{K - (M - w)^2}{8uvw}\right)\right) dM. \quad (1)$$

Read the integrand as: the density of M , times the fraction of W 's sphere in the cap. The min is the case of a full cap (W is always bad), the max the case of an empty one.

3 The three angles

Order the radii $a \geq b \geq c > 0$. In each of the three integrals (1) the clipping happens at simple places, and what is left is a polynomial in M .

The smallest radius, $W = C$. Here $K = a^2 + b^2 - c^2$; write $H = \sqrt{K}$. The cap is full for $M \leq H - c$, partial for $H - c \leq M \leq H + c$, and empty beyond. These points lie in $[a - b, a + b]$, because $K - (a - b + c)^2 = 2(a + c)(b - c) \geq 0$ and $(a + b - c)^2 - K = 2(a - c)(b - c) \geq 0$. Integrating the two polynomial pieces,

$$p_C = \frac{(H - c)^2 - (a - b)^2}{4ab} + \frac{2cK - \frac{1}{3}(H^3 - (H - 2c)^3)}{8abc} = \frac{1}{2} - \frac{c^2}{3ab}.$$

The middle radius, $W = B$. Here $K = a^2 + c^2 - b^2$ and the cap is partial on the whole range $[a - c, a + c]$: the numerator $K - (M - b)^2$ is a concave quadratic in M and is nonnegative at both ends, where it equals $2(a - b)(b + c)$ and $2(a - b)(b - c)$, and the full-cap condition $M + b \leq \sqrt{K}$ fails since $(a - c + b)^2 - K = 2(a + b)(b - c) \geq 0$. So

$$p_B = \int_{a-c}^{a+c} \frac{K - (M - b)^2}{8abc} dM = \frac{a - b}{2a} + \frac{c^2}{6ab}.$$

The largest radius, $W = A$. Here $K = b^2 + c^2 - a^2$. If $K \leq 0$ the cap is always empty and $p_A = 0$. Otherwise $h = \sqrt{K} \leq c$, the cap is partial for $|M - a| < h$ and empty elsewhere, and $[a - h, a + h] \subseteq [b - c, b + c]$ because $(a - b + c)^2 - h^2 = 2(a + c)(a - b) \geq 0$ and $(b + c - a)^2 - h^2 = 2(a - b)(a - c) \geq 0$. So

$$p_A = \int_{a-h}^{a+h} \frac{h^2 - (M - a)^2}{8abc} dM = \frac{h^3}{6abc},$$

which is also right when $K \leq 0$, with $h = 0$.

A triangle has at most one obtuse angle, and a right angle has probability zero, so $P = 1 - p_A - p_B - p_C$, which is the formula of Theorem 1.

4 The bound

Proof of Theorem 1. Fix $b \geq c > 0$ and let the largest radius a grow from b . Put

$$G(a) = 6abc\left(\frac{1}{2} - P\right) = 3bc(a - b) - c^3 + h^3.$$

At $a = b$ we have $h = c$ and $G(b) = 0$. While $b^2 + c^2 > a^2$, $h^3 = (b^2 + c^2 - a^2)^{3/2}$ and

$$G'(a) = 3(bc - ah), \quad b^2c^2 - a^2h^2 = b^2c^2 - a^2(b^2 + c^2 - a^2) = (a^2 - b^2)(a^2 - c^2),$$

which is positive for $a > b$; so G is strictly increasing there. Once $a^2 \geq b^2 + c^2$, $h = 0$ and G grows linearly with slope $3bc$. Hence $G(a) > 0$ for $a > b$: $P < \frac{1}{2}$ unless $a = b$, and $P = \frac{1}{2}$ when $a = b$. $\square$

Proof of Corollary 1. A rotationally invariant random point is $R\Theta$ with Θ uniform on the unit sphere and independent of the radius $R \geq 0$. Condition on the three radii. If all are positive,

Theorem 1 gives at most $\frac{1}{2}$. If exactly one is 0, say the triangle is $OA'B'$ with $|A'| = a \geq |B'| = b > 0$, then it is acute exactly when $0 < A' \cdot B' < b^2$, which has probability $b/(2a) \leq \frac{1}{2}$ because $A' \cdot B'/(ab)$ is uniform on $[-1, 1]$. If two are 0 the triangle is degenerate. So the probability is at most $\frac{1}{2}$. Equality needs equality almost surely in every case. If R is not almost surely one positive constant, choose a level r with $\Pr(R > r)$ and $\Pr(R \leq r)$ both positive; with positive probability exactly one of the three radii exceeds r , so the largest radius is unique and the conditional probability is below $\frac{1}{2}$ (or, with two radii at 0, it is 0). $\square$

Remark. Theorem 1 also gives the ball's value independently of Hall: with radial density $3r^2$, scale the largest of the three radii to 1, so the other two have joint density $18b^2c^2$ on $0 < c < b < 1$; integrating the formula gives $\frac{18}{35} - \frac{3}{70} = \frac{33}{70}$.

5 The plane is different

Put one vertex at the centre O and two uniform on the circle or sphere. The angles at the two boundary points are acute, since $(O - U) \cdot (V - U) = 1 - U \cdot V > 0$, and the angle at O is acute exactly when $U \cdot V > 0$, which has probability $\frac{1}{2}$ by the symmetry $V \mapsto -V$. So this triangle is acute half the time in either dimension. Now let each point be the centre with probability ε and uniform on the boundary otherwise. No centre gives the boundary value, one centre gives $\frac{1}{2}$, and two or three give a degenerate triangle:

$$\begin{aligned} F_2(\varepsilon) &= \frac{(1 - \varepsilon)^3}{4} + \frac{3\varepsilon(1 - \varepsilon)^2}{2} = \frac{(1 - \varepsilon)^2(1 + 5\varepsilon)}{4} && \text{(plane),} \\ F_3(\varepsilon) &= \frac{(1 - \varepsilon)^3}{2} + \frac{3\varepsilon(1 - \varepsilon)^2}{2} = \frac{(1 - \varepsilon)^2(1 + 2\varepsilon)}{2} && \text{(space).} \end{aligned}$$

In the plane $F_2'(0) = \frac{3}{4} > 0$, and for example $F_2(\frac{1}{10}) = \frac{243}{800} > \frac{1}{4}$: the circle is not even a local best. In space $F_3(\varepsilon) = \frac{1}{2} - \frac{3}{2}\varepsilon^2 + \varepsilon^3 < \frac{1}{2}$ for $0 < \varepsilon \leq 1$, as Corollary 1 requires. The difference is the boundary value: a triangle with a vertex at the centre is acute half the time in both dimensions, which is better than the circle's $\frac{1}{4}$ and no better than the sphere's $\frac{1}{2}$.

Remark. This does not prove that the uniform disc beats the circle: the disc spreads its mass over all radii, not just the centre, and very unequal radii can hurt in the plane too. A direct geometric reason for the disc's gain is still missing. The wider question, which distribution in space maximises the probability without assuming rotational invariance, is open; for uniform points in convex bodies Hall conjectured that the ball is best, and Khan showed it is a local maximum [1, 2]. Corollary 1 answers the rotationally invariant case of the question asked on Mathematics Stack Exchange [4].

Verification. Lemma 1, the clipped uniform tail and the form of the integrand in (1), the three integrals p_A, p_B, p_C for every $a \geq b \geq c > 0$, the formula and the bound of Theorem 1 with its equality case, the one-centre case $b/(2a) \leq \frac{1}{2}$ and the two mixture polynomials are checked in Lean 4 with Mathlib [5], using only the standard axioms. Quoted rather than formalised: the hat-box theorem and the conditioning that turns it into (1), and the decomposition of a rotationally invariant law used in the proof of Corollary 1. A Python program checks the three cap probabilities against direct numerical integration of the defining integrals for 43 radius triples, and the mixture and ball arithmetic exactly; six deliberately wrong variants fail it. An independent program, sharing no code with it, estimates $P(a, b, c)$, the ball value and the mixtures by simulating actual points, and integrates the formula exactly against the ball's radial density to get $\frac{33}{70}$. Everything is at <https://github.com/jvbk/mathematics/tree/main/why-the-sphere-beats-the-ball>.

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Vamshi Jandhyala, London, UK.